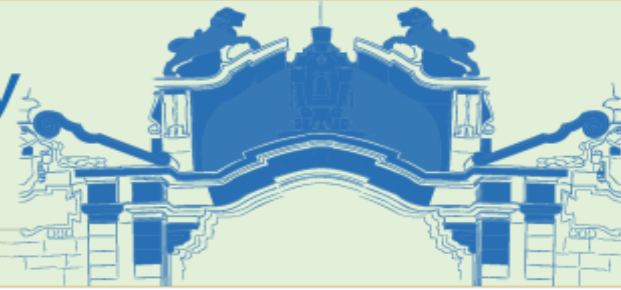




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Addis Ababa Institute of Technology

School of Mechanical and Industrial Engineering

Engineering Materials I
(MEng 2122)

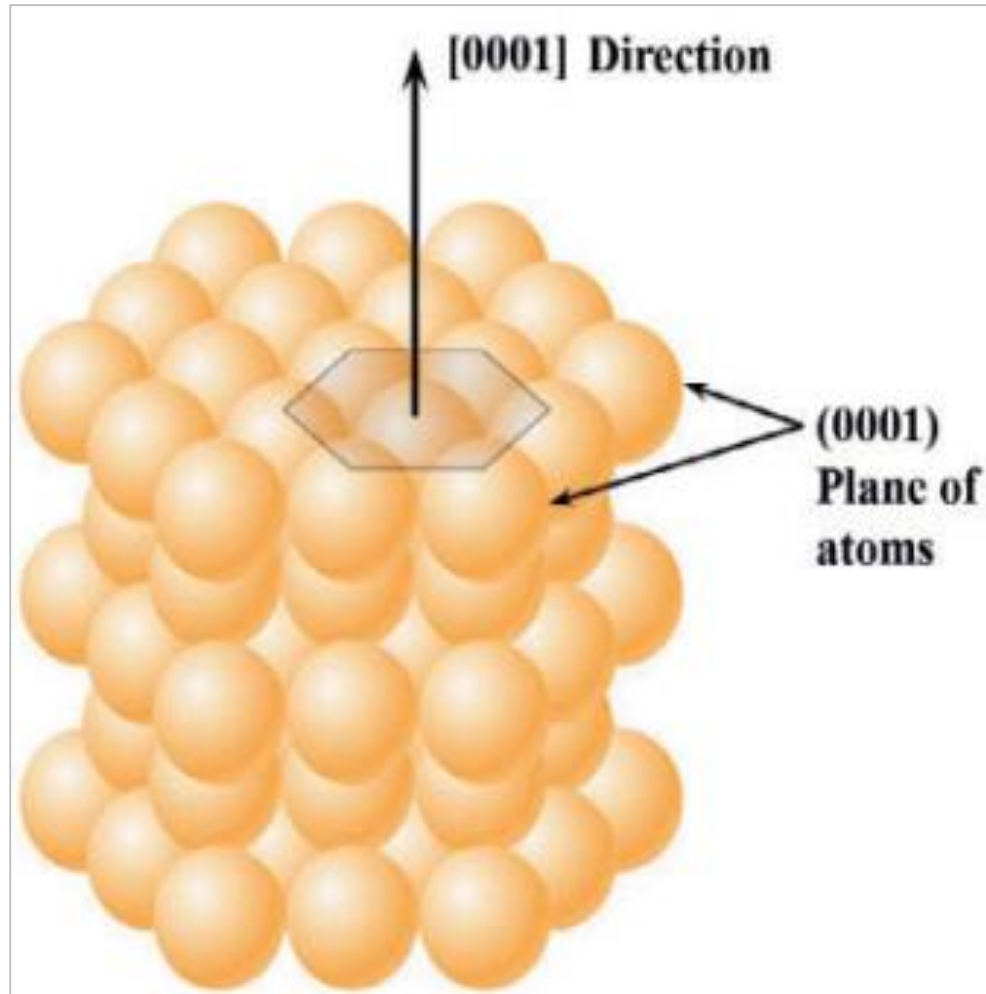
*Compiled by: Ayalsew Dagnew
(MSc. Aerospace Eng`g)*

June, 2022

Chapter - 3

Structure of Crystalline Solids

Structure of Crystalline Solids



Content Outline:

- Introduction to Crystal Structure
- Crystal Structures
- Crystallographic Points, Directions & Planes
- Crystalline Materials
- Non-Crystalline Materials

Why we Study the Structure of Crystalline Solids?

- To understand structure of solid materials.
- To identify the difference between crystalline and noncrystalline materials.
- To come up with the concept, properties of some materials directly related to their crystal structures.
- Understand the significant property differences exist between crystalline and noncrystalline materials having the same composition.
- Therefore, this intends to study the structural properties of solids.

Learning Objectives

After completing this chapter, you should be able to:

- Describe the difference in atomic/molecular structure b/n crystalline and noncrystalline materials.
- Draw unit cells for FCC, BCC, and HCP crystal structures.
- Derive the relationships b/n unit cell edge length and atomic radius for FCC, BCC and HCP crystal structures.
- Compute the densities for metals having FCC and BCC crystal structures given their unit cell dimensions.
- Sketch the direction corresponding to given three direction indexes in a unit cell.
- Specify the Miller indices for a plane that has been drawn within a unit cell.
- Describe how FCC and HCP crystal structures may be generated by the stacking of close-packed planes of atoms.
- Distinguish b/n single crystals and polycrystalline materials.
- Define *isotropy* and *anisotropy*.

Introduction to Crystal Structure

Crystal structure:

- The present discussion is devoted to the **next level** of the structure of materials, specifically, to some of the **arrangements** that may be assumed by **atoms** in the solid state. Within this framework, concepts of **crystallinity** and **noncrystallinity** are introduced.
- **Crystallinity**: refers to the degree of structural order in a solid, and the atoms or molecules are arranged in a **regular periodic** manner.
- The *degree of crystallinity* has a big influence on **hardness**, **density**, **transparency** and **diffusion**.
- **Non-crystallinity**: also called **amorphous**, is a solid that lacks the **long-range** order that is characteristic of a crystal.

Introduction to Crystal Structure

- The *three common crystal structures* found in metals are expressed by **crystallographic points**, directions, and planes.
- A **crystalline solid** can either be a **single crystalline** or a **polycrystalline**.
- The **determination of crystal structures** experimentally using **x-ray diffraction** techniques are also studied.



Crystallinity increases



Amorphous

Crystallinity

Crystal Structures

Fundamental Concepts

- **Solid materials** may be classified according to the regularity with which atoms or ions are arranged with respect to one another.
- A **crystalline** material is one in which the atoms are **arranged** in a **repeating** or periodic array over large atomic distances.
- Under **normal solidification** conditions the following materials can form **crystalline structures**.
 - all metals,
 - many ceramics, and
 - certain polymers.
- Those that **do not crystallize** are called **noncrystalline** materials. They also called **amorphous** materials.
- In **noncrystalline** materials the long-range **atomic order** is absent.

Crystal Structures

Fundamental Concepts

- Properties of *crystalline solids* depend on the *crystal structure* of the material.

Which means affecting by the atoms, ions, or molecules *arrangement*.

- There are extremely large number of *different crystal structures* having long-range atomic order.
- These *crystal structures* vary from *simple* to *complex* materials.
 - The simplest crystal structures are belong to → *metals*
 - The complex crystal structures are refers to → *ceramics* and *polymers*

Crystal Structures

Fundamental Concepts

- When *crystalline structures* are described, *atoms* or *ions* are considered as being solid spheres having well-defined diameters.
- This is termed the *atomic hard-sphere model* in which spheres representing nearest-neighbor atoms touch one another. [refer to Figure 3.1c]
- Sometimes the term *lattice* is used in the context of *crystal structures*.
- *Lattice* is a *three-dimensional array* of points coinciding with atom positions or sphere centers.

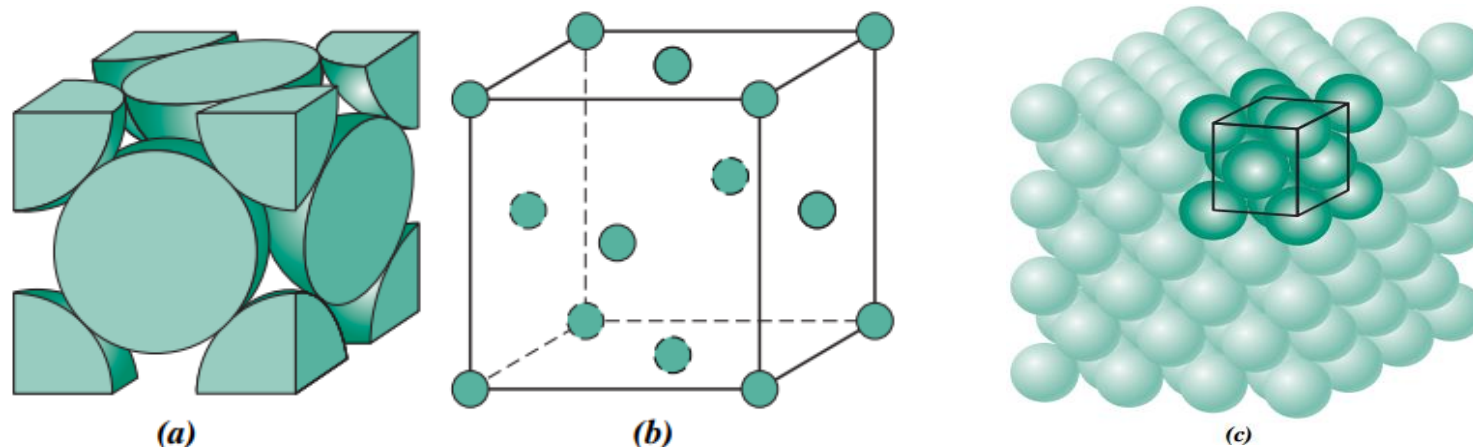


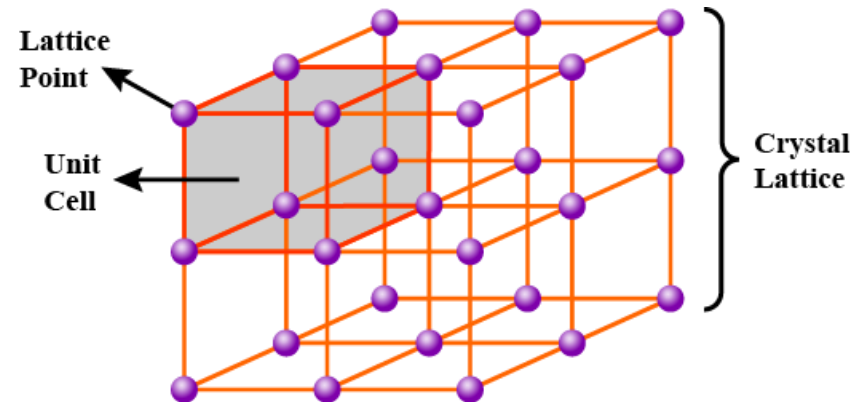
Figure 3.1 For the face-centered cubic crystal structure, (a) a hard-sphere unit cell representation, (b) a reduced-sphere unit cell, and (c) an aggregate of many atoms.

Crystal Structures

Unit Cell

What is a unit cell?

- A **unit cell** is the basic & smallest portion of a crystal lattice that shows the **three-dimensional pattern** of the entire crystal.
 - A **crystal** can be thought of as the same **unit cell** repeated over & over in three dimensions.
 - Atoms of **crystalline solids** forms – a **repetitive pattern**.
 - These repetitive substructure entities are called **unit cells**.
- A **unit cell** is chosen to represent the symmetry of the crystal structure.
 - The **unit cell** is the basic structural unit or building block of the **crystal structure**.
 - A single **unit cell** may be chosen for a particular crystal structure.



Crystal Structures

Metallic Crystal Structures

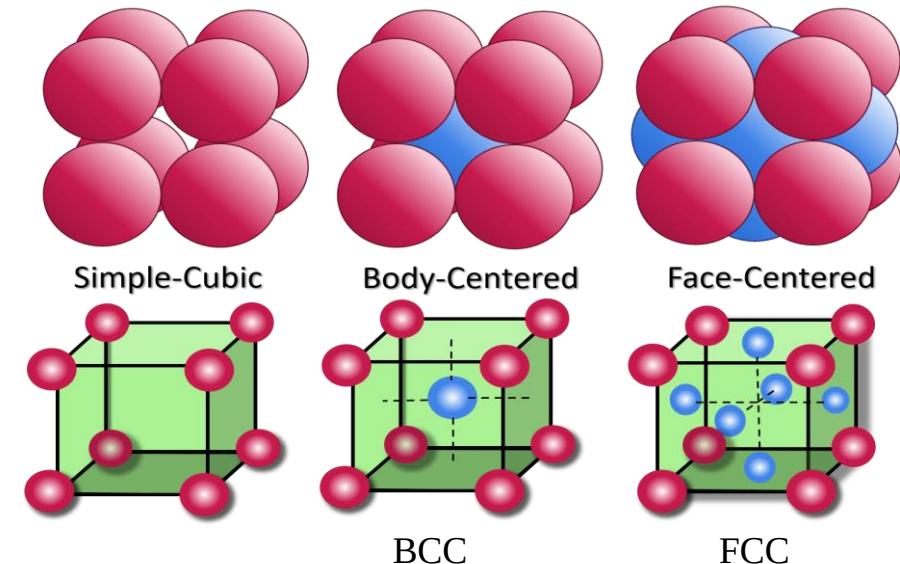
What is metallic crystal structure?

Metallic crystal is a crystalline solid in which the atoms are held together by metallic bonds.

- The **atomic bonding** in this group of materials is metallic and thus nondirectional in nature.
- Most **metallic crystal structures** are *dense atomic packing*.
- **Three simple crystal structures** are found for most of the common metals:
 - Face-centered cubic /FCC/,
 - Body-centered cubic /BCC/, and
 - Hexagonal close-packed /HCP/.

The **cubic crystal system** is a crystal system where the unit cell is in the shape of a cube. It is one of the most common and simplest shapes found in crystals and minerals.

Types of Unit Cells

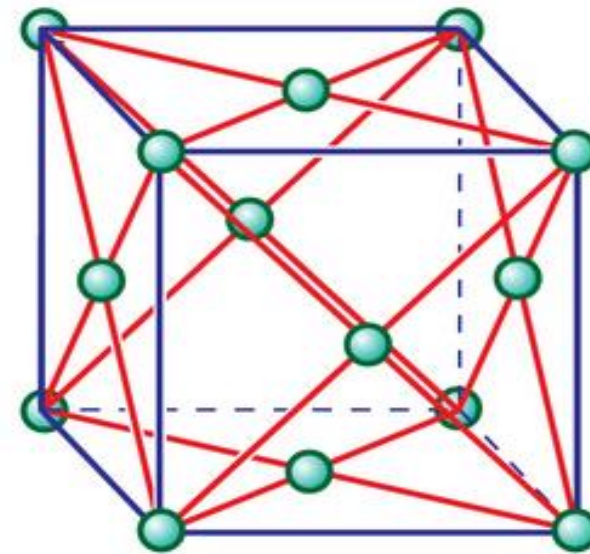


Crystal Structures

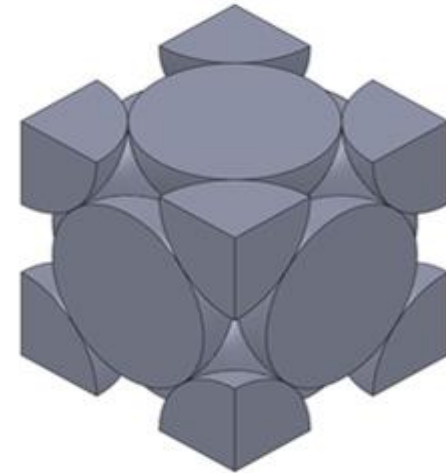
Metallic Crystal Structures

FCC Crystal Structure:

- Crystal structures of many metals has a unit cell of cubic geometry.
- Atoms are located at each corners and six cube faces.
- Some of the metals having FCC crystal structure are: Cu, Al, Ag, and Au. [see Table 3.1]
- FCC unit cell consists of atoms representing as spheres.
- These spheres or ion cores touch one another across a face diagonal.



face-centred cubic (fcc)



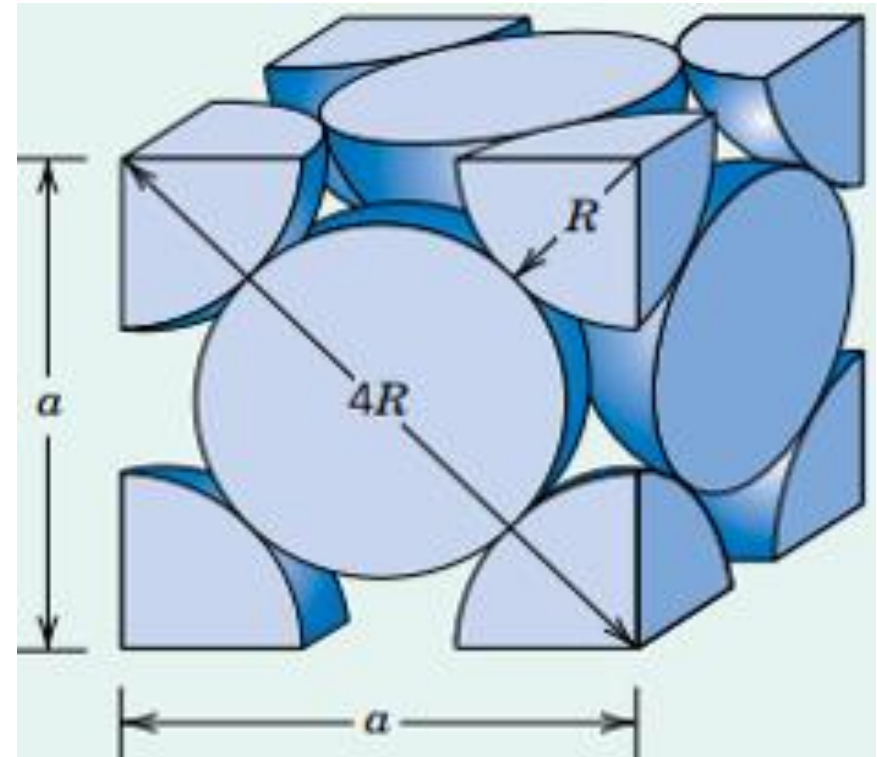
Crystal Structures

Metallic Crystal Structures

- The cube **edge length** a and the **atomic radius** R are related through.
- **Unit cell edge length** for FCC;

$$a = 2R\sqrt{2}$$

- Metals with FCC crystal structure;
Al, Cu, Au, Pb, Ni, Pt, and Ag (*refer Table 3.1*).



Refer to [Figure 3.1](#)

Crystal Structures

Metallic Crystal Structures

- The *number of atoms* per *unit cell* (N) can be computed using the following formula:

$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8}$$

Where; N_i - the number of interior atoms

N_f - the number of face atoms

N_c - the number of corner atoms.

- For the *FCC* crystal structure: $N_c = 8$, $N_f = 6$, and $N_i = 0$.

Thus, from the above equation;

$$N = 0 + \frac{6}{2} + \frac{8}{8} = 4$$

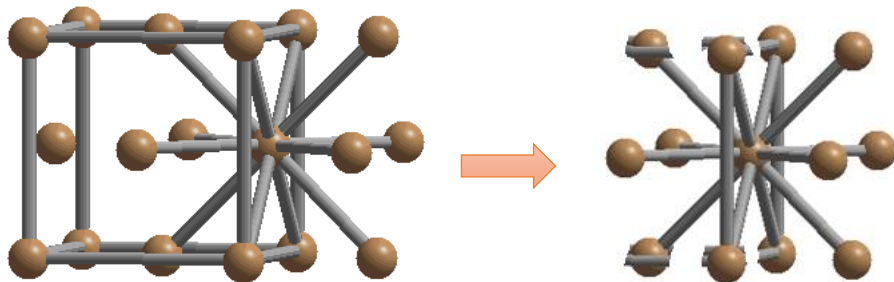
- A total of *four* full atoms are assigned to a given FCC unit cell.
- *Corner* and *face* positions are really equivalent.
- *Two* important *characteristics* of a crystal structure are the;
 - *coordination number* and
 - *atomic packing factor (APF)*.
- For metals, each atom has the *same number of nearest-neighbor* or touching atoms, which is the *coordination number*.

Crystal Structures

Metallic Crystal Structures

What is coordination number?

- For **molecules** and **polyatomic** ions the **coordination number** of an atom is determined by simply counting the other atoms to which it is bonded.
- For FCC, the coordination number is **12**.



Coordination number

- The **APF** is the sum of the sphere **volumes of all atoms** within a unit cell divided by the **unit cell volume**.

$$\text{APF} = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}}$$

$$V = \frac{4}{3}\pi r^3$$

- In FCC structure, the APF is 0.74, which is the maximum packing possible for spheres all having the same diameter.
- Metals** have relatively **large** APFs to maximize the shielding provided by the free electron cloud.

Crystal Structures

Metallic Crystal Structures

- ❖ Refer to the following link for further understanding about FCC crystal structure and follow the videos in the link thread ...:

<https://www.youtube.com/watch?v=GSPVC34ijIA>

EXAMPLE PROBLEM 3.2

Computation of the Atomic Packing Factor for FCC

Show that the atomic packing factor for the FCC crystal structure is 0.74.

Solution

The APF is defined as the fraction of solid sphere volume in a unit cell, or

$$\text{APF} = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}} = \frac{V_S}{V_C}$$

Both the total atom and unit cell volumes may be calculated in terms of the atomic radius R . The volume for a sphere is $\frac{4}{3}\pi R^3$, and because there are four atoms per FCC unit cell, the total FCC atom (or sphere) volume is

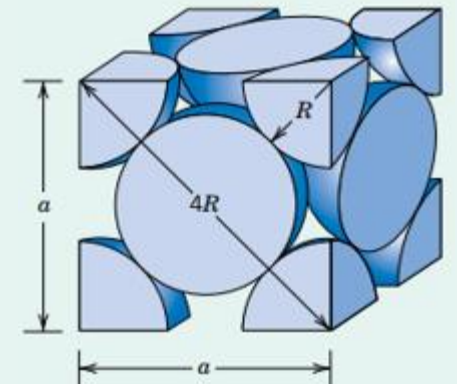
$$V_S = (4)\frac{4}{3}\pi R^3 = \frac{16}{3}\pi R^3$$

From Example Problem 3.1, the total unit cell volume is

$$V_C = 16R^3\sqrt{2}$$

Therefore, the atomic packing factor is

$$\text{APF} = \frac{V_S}{V_C} = \frac{\left(\frac{16}{3}\right)\pi R^3}{16R^3\sqrt{2}} = 0.74$$

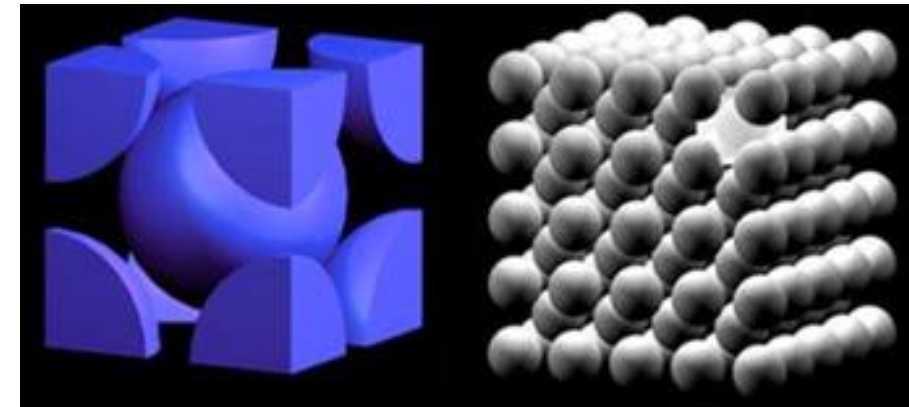
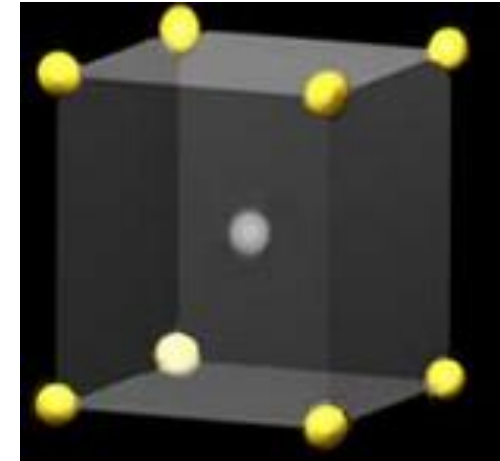


Crystal Structures

Metallic Crystal Structures

BCC Crystal Structure:

- The **BCC** is another common metallic crystal structure.
- It has a **cubic unit cell** with atoms located at all *eight corners* and a *single* atom at the cube *center*.
- The atoms represented by *hard-sphere* and *reduced-sphere* models, respectively shown in *Figure 3.2a & b*.



Crystal Structures

Metallic Crystal Structures

- *Center* and *corner* atoms touch one another along cube *diagonals*,

and unit cell length a and atomic radius R are related through;

$$a = \frac{4R}{\sqrt{3}}$$

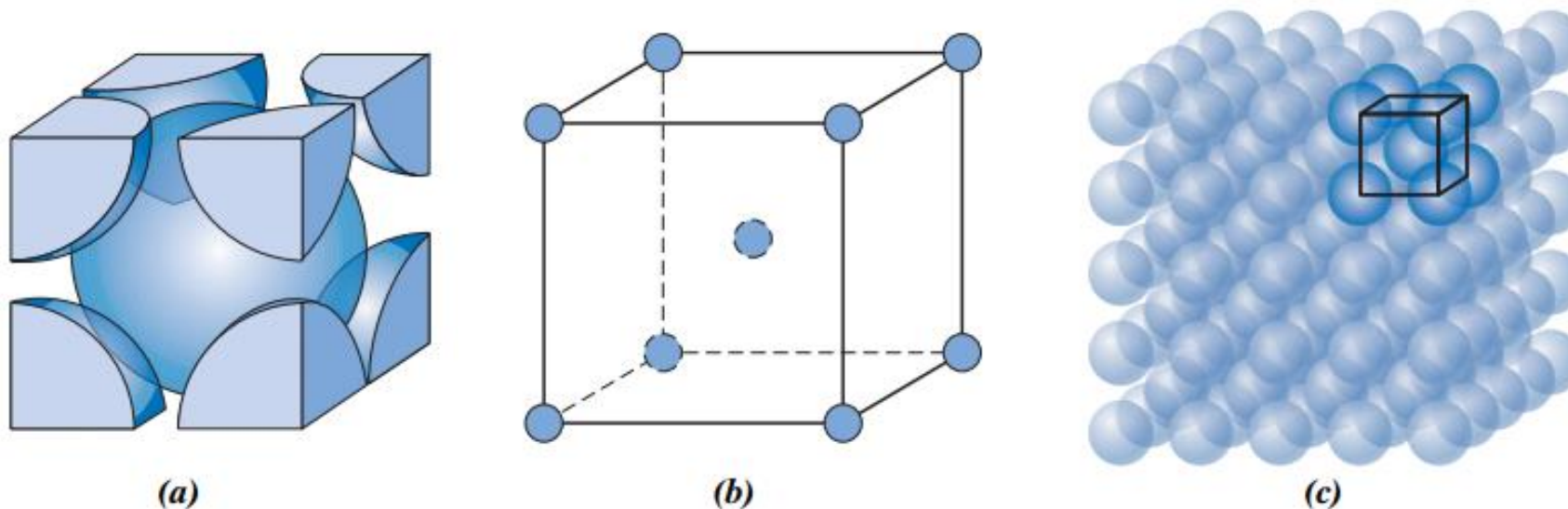


Figure 3.2 For the body-centered cubic crystal structure, (a) a hard-sphere unit cell representation, (b) a reduced-sphere unit cell, and (c) an aggregate of many atoms.

❖ Refer to the following link for further understanding about BCC crystal structure and follow the videos in the link thread ...:

https://www.youtube.com/watch?v=_9RnbGqtkd4

Crystal Structures

Metallic Crystal Structures

- The number of atoms per BCC unit cell is;

$$\begin{aligned} N &= N_i + \frac{N_f}{2} + \frac{N_c}{8} \\ &= 1 + 0 + \frac{8}{8} = 2 \end{aligned}$$

- The **coordination number** for the BCC crystal structure is 8.
- The coordination number is less for BCC than for FCC.
- The **atomic packing factor** is also lower for BCC, which is 0.68.

- It is also possible to have a unit cell that consists of atoms situated *only* at the *corners* of a cube.
 - This is called the **simple cubic** (SC) *crystal structure*; hard-sphere and reduced-sphere models as shown in *Figure 3.3a & b*.
- **None** of the **metallic** elements have the **simple cubic crystal structure** because of its relatively low APF.
- The **only simple-cubic** element is **polonium**, which is considered to be a metalloid.

Crystal Structures

Metallic Crystal Structures

HCP Crystal Structure:

What is hexagonal close-packed crystal structure?

- Not all **metals** have unit cells with cubic symmetry.
- The final common metallic crystal structure is **hexagonal**.
- A reduced-sphere unit cell for **HCP**, *Figure 3.4a* shows.
- The **number of atoms** per **unit cell** for the **HCP** crystal structure is, N ;

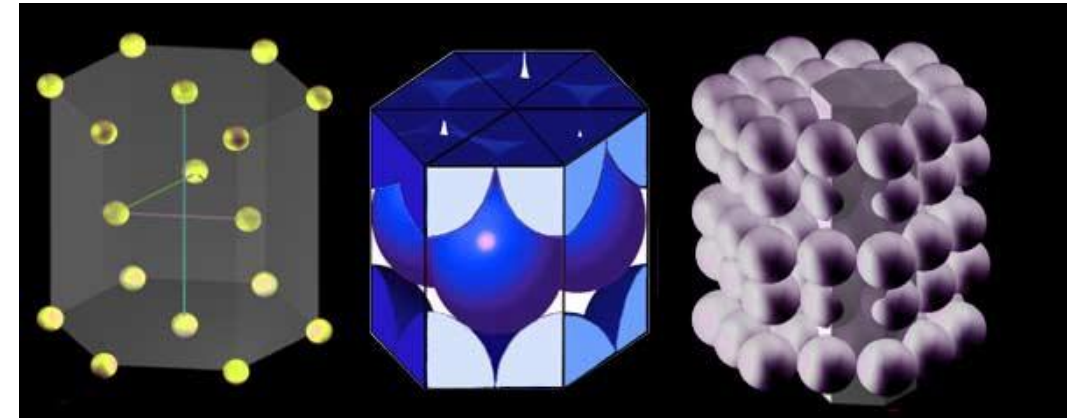


Figure a: hexagonal close-packed unit cell.

❖ Refer to see about HCP:

<https://www.youtube.com/watch?v= OhQv5gsoSQ>
<https://www.youtube.com/watch?v=WuclTFbINq4>

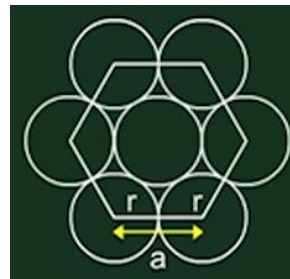
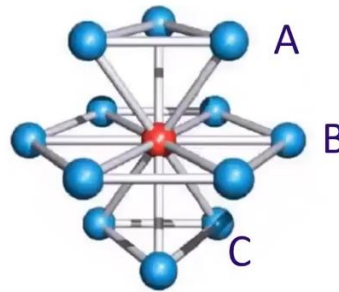
Crystal Structures

Metallic Crystal Structures

- The value of N for HCP is found,

$$N = 3 + \frac{2}{2} + \frac{12}{6} = 6$$

- Thus, in HCP 6 atoms are assigned to each unit cell.
- The *coordination number* of HCP is 12.
- The *APF* for HCP is 0.74.
- The *coordination number* and *APF* of FCC & HCP are the same.
- The *HCP metals* include cadmium, Mg, Ti, and Zn. Some of these listed in Table 3.1.



What is the real shape of HCP unit cell?

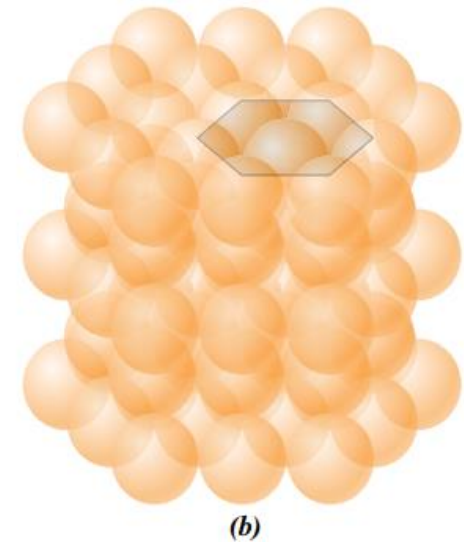
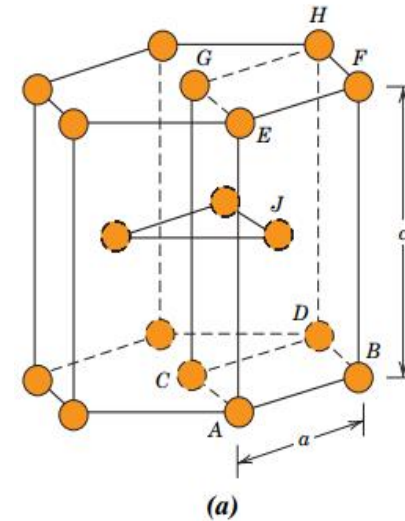


Figure 3.4 For the hexagonal close-packed crystal structure, (a) a reduced-sphere unit cell (a and c represent the short and long edge lengths, respectively), and (b) an aggregate of many atoms.

- *What is c in terms of the radius R ?*
- *Show the APF of HCP is 0.74.*

Crystal Structures

Metallic Crystal Structures

Table 3.1: Atomic radii and crystal structure for 16 metals.

<i>Metal</i>	<i>Crystal Structure^a</i>	<i>Atomic Radius^b (nm)</i>	<i>Metal</i>	<i>Crystal Structure</i>	<i>Atomic Radius (nm)</i>
Aluminum	FCC	0.1431	Molybdenum	BCC	0.1363
Cadmium	HCP	0.1490	Nickel	FCC	0.1246
Chromium	BCC	0.1249	Platinum	FCC	0.1387
Cobalt	HCP	0.1253	Silver	FCC	0.1445
Copper	FCC	0.1278	Tantalum	BCC	0.1430
Gold	FCC	0.1442	Titanium (α)	HCP	0.1445
Iron (α)	BCC	0.1241	Tungsten	BCC	0.1371
Lead	FCC	0.1750	Zinc	HCP	0.1332

^aFCC = face-centered cubic; HCP = hexagonal close-packed; BCC = body-centered cubic.

^bA nanometer (nm) equals 10^{-9} m; to convert from nanometers to angstrom units ($\text{\AA}$), multiply the nanometer value by 10.

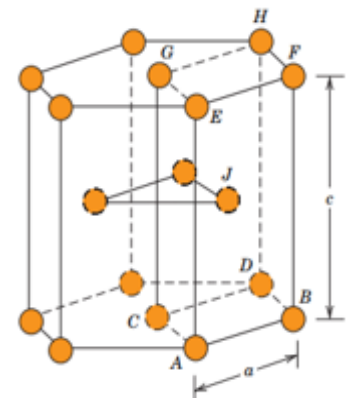
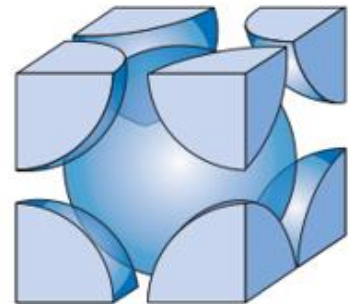
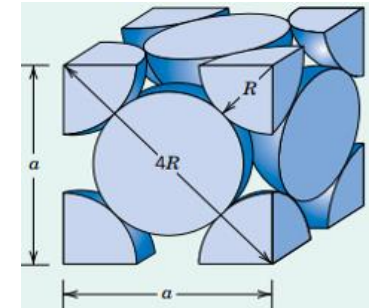


Figure: common metallic crystal structures.

Crystal Structures

Density Calculations

What is density?

Density is mass of a unit volume of a material substance.

Theoretical density for metals:

$$\rho = \frac{nA}{V_C N_A}$$

where,

n = number of atoms associated with each unit cell

A = atomic weight

V_C = volume of the unit cell

N_A = Avogadro's number (6.022×10^{23} atoms/mol)

Example: Copper has an atomic radius of 0.128nm with FCC crystal structure and an atomic weight of 63.5g/mol. Determine its theoretical density and compare the answer with its measured density, 8.94 g/cm³.